FAKE NEWS BUSTER

Fake and Real News Classification using Machine Learning

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Project Report

1. Introduction

Fake news is false or misleading information presented as news. It can spread quickly through websites and social media and may confuse readers. The aim of this project is to build a simple machine learning model that classifies a news article as Fake News or Real News based on its text.

This project uses Python and common data-science libraries. Text data is converted into numerical features so that a machine learning algorithm can process it. The model used is Multinomial Naive Bayes, which is commonly used for text classification tasks.

2. Objectives

The main objectives of the project are to load a labelled news dataset, inspect and clean the data, explore patterns in the text, convert text into numerical features, train and compare Multinomial Naive Bayes models, and test the selected model on new examples.

3. Dataset Description

The project uses two CSV files: Fake.csv and True.csv. The Fake.csv records are assigned label 1, and the True.csv records are assigned label 0. The two datasets are combined into one DataFrame for analysis.

The original combined dataset contains 44,898 rows before cleaning: 23,481 fake-news records and 21,417 real-news records. The columns include title, text, subject, date, and label. The title and text are used as the main text content, while label is the target column.

4. Tools and Libraries

Python was used in a notebook environment. Pandas was used to load and manage the dataset; NumPy was used for numerical operations; Matplotlib and Seaborn were used to create charts; and scikit-learn was used for text vectorization, splitting the data, training the model, and evaluating predictions.

5. Data Understanding and Cleaning

The dataset was inspected using info(), describe(), missing-value counts, and duplicate counts. This helped identify the columns, data types, and general quality of the data.

Duplicate rows were removed. Missing title and text values were filled with empty strings and converted to text. A Reuters-style tag at the beginning of some text entries was removed. The subject and date columns were not used as model features, helping the model focus on the news content rather than metadata. The label column was retained as the target.

6. Exploratory Data Analysis (EDA)

Charts were created to understand the dataset before model training. A count plot showed the number of real and fake news records. A box plot compared text length across the two classes. Word-frequency charts showed common words in fake and real news after simple filtering.

The class counts show that the original dataset contains both classes, with more fake-news records than real-news records. Text length and frequent-word patterns provide useful descriptive information, but these patterns alone do not prove that an individual article is true or false.

7. Text Vectorization and Feature Selection

Machine learning models cannot directly process raw sentences, so the title and article text were combined into one text field. CountVectorizer converted the combined text into a numerical word-count matrix. English stop words were removed and the vocabulary was limited to 3,000 features to keep the feature space manageable.

LabelEncoder was used to prepare the target labels. The chi-square test was applied to score words according to how strongly they were associated with the target classes. The highest-scoring words were inspected to understand which terms were most informative for classification.

8. Train-Test Split and Model

The data was split into training and testing sets using an 80:20 ratio. Stratification was used so that the class proportions were maintained in both sets, and random_state=42 was used to make the split reproducible.

The selected algorithm was Multinomial Naive Bayes. It estimates class probabilities from word counts and applies Bayes' theorem to predict a class. The algorithm assumes that features are conditionally independent given the class. Real words can be related to one another, so this assumption is a simplification of natural language.

9. Model Evaluation and Tuning

The baseline model was evaluated using Accuracy, Precision, Recall, F1 score, Specificity, AUC, a confusion matrix, and a classification report. These measures provide more information than accuracy alone. Precision measures how often fake-news predictions are correct; recall measures how many actual fake-news items are identified; F1 combines precision and recall; specificity measures correct identification of real-news items; and AUC measures class separation across thresholds.

Three smoothing settings were compared: alpha = 0.1, alpha = 0.5, and alpha = 1.0. The results table was sorted by F1 score, and a bar chart compared Accuracy, Precision, Recall, and F1. The final model was selected using the highest F1 score among the tested settings.

10. Results

The exact evaluation values should be copied from the final output of the notebook. They are left blank below so the report reflects the actual experiment rather than estimated or invented scores.

Metric	Value from notebook
Best alpha	_____
Accuracy	_____
Precision	_____
Recall	_____
F1 score	_____
Specificity	_____
AUC	_____

11. Testing with New News Examples

Three new example sentences were passed through the same fitted CountVectorizer using transform(), and the final trained model predicted a label for each one. Using the same fitted vectorizer is important because the new text must use the same vocabulary and feature order as the training data.

These outputs are model predictions, not proof of the truth of the news. A prediction may be incorrect, especially when a new article uses wording or topics that differ from the training dataset. Important claims should be verified using reliable sources.

12. Limitations

The model learns patterns from the dataset and may be affected by bias, outdated content, repeated stories, and differences in writing style. Word counts do not fully understand context, sarcasm, or the meaning of an article. A high score on this dataset does not guarantee correct predictions on every real-world news item.

13. Conclusion

In this project, a basic fake-news classification pipeline was developed using Python and Multinomial Naive Bayes. The work included dataset loading, inspection, cleaning, exploratory charts, text vectorization, feature selection, model training, evaluation, and comparison of alpha settings. The final model was then used to predict labels for three new text examples.

The project demonstrates how machine learning can support text classification. However, the model should be treated as a screening tool rather than a final authority on whether news is true. Future improvements could include testing on newer data, using cross-validation, checking data leakage carefully, and comparing additional text-classification algorithms.

1. Fake.csv and True.csv dataset files supplied for this project.

2. Scikit-learn documentation: https://scikit-learn.org/

3. Pandas documentation: https://pandas.pydata.org/docs/

4. Matplotlib and Seaborn documentation: https://matplotlib.org/ and https://seaborn.pydata.org/

Before submission, fill the Results table using the actual values printed by your notebook.